

```

1  package Cdc;
2
3
4  import java.io.*;
5  import javax.swing.*;
6  import java.awt.*;
7  import java.util.*;    //for Vector
8  import java.lang.*;
9  import java.lang.*;
10 import java.awt.event.*;
11 import udpbc.*;
12
13
14
15 /**
16  * <p>Title: </p>
17  * <p>Description: </p>
18  * <p>Copyright: Copyright (c) 2005</p>
19  * <p>Company: </p>
20  * @author not attributable
21  * @version 1.0
22  */
23
24
25
26 public class Cdc {
27     static UDPClient ul;
28     // private int[][] characterDataBase = new
29     int[characterNumber][characterDataBaseElement];
30     //characterDataBaseElement: speed, 生命初始值, 攻擊使別人失血量。
31     //          0          1          2
32     public int clientEatNPC = 0;
33     public int[][] characterDataBase = {
34         {
35             0, 250, 20
36         }, {
37             0, 200, 30
38         }, {
39             0, 220, 25
40         }, {
41             0, 200, 25
42         }
43     };
44     int updatearray[];
45
46     //npc: x, y, eaten, index(clientNo = nowCharacterNum+1)。
47     //      0    1    2    3
48     int[] npc = {
49         3, 6, 0, 0
50     };
51     int npcElement = npc.length;
52
53     boolean eaten = false;
54     static final int characterNumber = 6; //clientNo不用0, 從1開始。
55     int nowCharacterNum = 0;
56     static final int characterElement = 7;
57     //characterElement: 動物種類, dir, speed, x, y, 生命值, (是1) (否0)發動攻擊。
58     //          0    1    2    3    4    5    6
59     static public int[][] character = new int[characterNumber][characterElement];
60     static final int MAX_map = 30;
61     static public int[][] map = new int[MAX_map][MAX_map];
62     Vector info = new Vector();
63     private int itemNumber = 1; //info vector 第0個為itemNumber, 目前沒有用到。
64     static public NPC n1;
65     static position p1;
66     // DiedJudge dl;
67
68     public Cdc() {
69         n1 = new NPC(this);
70         p1 = new position(this);
71         // dl = new DiedJudge(this);
72         // dl.start();
73         p1.start();
74         // n1.start();
75     }
76
77
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73 map[0][13] = map[0][14] = map[0][15] = map[0][16] = -1;
74 map[1][13] = map[1][14] = map[1][15] = map[1][16] = -1;
75 map[2][13] = map[2][14] = map[2][15] = map[2][16] = -1;
76 map[10][8] = map[10][9] = map[10][10] = map[10][11] = -1;
77 map[10][12] = map[10][17] = map[10][18] = map[10][19] = -1;
78 map[10][20] = map[10][21] = -1;
79 map[11][8] = map[11][9] = map[11][10] = map[11][11] = -1;
80 map[11][12] = map[11][17] = map[11][18] = map[11][19] = -1;
81 map[11][20] = map[11][21] = -1;
82 map[12][8] = map[12][9] = map[12][10] = map[12][11] = -1;
83 map[12][12] = map[12][17] = map[12][18] = map[12][19] = -1;
84 map[12][20] = map[12][21] = -1;
85 map[17][8] = map[17][9] = map[17][10] = map[17][11] = -1;
86 map[17][12] = map[17][17] = map[17][18] = map[17][19] = -1;
87 map[17][20] = map[17][21] = -1;
88 map[18][8] = map[18][9] = map[18][10] = map[18][11] = -1;
89 map[18][12] = map[18][17] = map[18][18] = map[18][19] = -1;
90 map[18][20] = map[18][21] = -1;
91 map[19][8] = map[19][9] = map[19][10] = map[19][11] = -1;
92 map[19][12] = map[19][17] = map[19][18] = map[19][19] = -1;
93 map[19][20] = map[19][21] = -1;
94 map[13][0] = map[13][1] = map[13][2] = map[13][27] = -1;
95 map[13][28] = map[13][29] = -1;
96 map[14][0] = map[14][1] = map[14][2] = map[14][27] = -1;
97 map[14][28] = map[14][29] = -1;
98 map[15][0] = map[15][1] = map[15][2] = map[15][27] = -1;
99 map[15][28] = map[15][29] = -1;
100 map[16][0] = map[16][1] = map[16][2] = map[16][27] = -1;
101 map[16][28] = map[16][29] = -1;
102 map[27][13] = map[27][14] = map[27][15] = map[27][16] = -1;
103 map[28][13] = map[28][14] = map[28][15] = map[28][16] = -1;
104 map[29][13] = map[29][14] = map[29][15] = map[29][16] = -1;
105 init();
106 }
107
108 public void init() {
109     for (int i = 0; i < characterNumber; i++)
110         for (int j = 0; j < characterElement; j++)
111             character[i][j] = -99;
112 }
113
114
115 public void addVirtualCharacter(int clientNo, int x, int y, int animalKind) {
116     nowCharacterNum++;
117     character[clientNo][0] = animalKind;
118     character[clientNo][1] = 0;
119     character[clientNo][2] = characterDataBase[animalKind][0];
120     character[clientNo][3] = x;
121     character[clientNo][4] = y;
122     character[clientNo][5] = characterDataBase[animalKind][1];
123     character[clientNo][6] = 0;
124
125     /* for(int i=y-1; i<= y+1; i++){
126         for(int j=x-1; j<=x+1; j++)
127             map[i][j] = clientNo;
128     }*/
129
130 //     System.out.println("get addVirtualCharacter");
131 }
132
133
134 public Vector getAddInfo() {
135     String infoString = "C";
136     info.add(Integer.toString(itemNumber)); //SET itemNumber;
137     for (int i = 1; i <= nowCharacterNum; i++) {
138         infoString = infoString + Integer.toString(i) +
139             Integer.toString(character[i][0]);
140         info.add(infoString);
141         infoString = "C";
142     }
143     infoString = "I0";
144     info.add(infoString);
145     return info;

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146     }
147
148     synchronized public Vector getUpdateInfo() {
149         // int itemNumber = 0; //info vector 第0個為itemNumber，目前沒有用到。
150         info.add(Integer.toString(itemNumber));
151         String infoString = "C";
152         String tempString = "";
153         for (int i = 1; i <= nowCharacterNum; i++) {
154             infoString = infoString + Integer.toString(i); //clientNo
155             for (int j = 1; j < characterElement; j++) {
156                 // System.out.print(character[i][j]);
157                 tempString = Integer.toString(character[i][j]);
158
159                 if (j == 2 ) { //2位數。
160                     if (tempString.length() == 1) //不足2位數補0。
161                         tempString = "0" + tempString;
162                 }
163                 else if (j == 3 || j == 4 || j == 5) { //3位數。
164                     if (tempString.length() != 3) {
165                         if (tempString.length() == 1)
166                             tempString = "00" + tempString;
167                         else
168                             tempString = "0" + tempString;
169                     }
170                 }
171                 infoString = infoString + tempString;
172                 // System.out.println(infoString);
173             }
174             // System.out.println("");
175             // System.out.println(infoString);
176             info.add(infoString);
177             infoString = "C";
178         }
179
180         // System.out.println("in getUpdateInfo. npcElement is " + npcElement );
181         infoString = "I";
182         for(int j=0; j<itemNumber; j++){ //把item的東西加到vector裡面。
183             for (int i = 0; i < npcElement-1; i++) {
184                 tempString = Integer.toString(npc[i]);
185                 if (i == 0 || i == 1) {
186                     if (tempString.length() != 3) {
187                         if (tempString.length() == 1)
188                             tempString = "00" + tempString;
189                         else
190                             tempString = "0" + tempString;
191                     }
192                 }
193                 infoString = infoString + tempString;
194             }
195             infoString = infoString + Integer.toString(0);
196             info.add(infoString);
197             infoString = "I";
198         }
199
200         return info;
201     }
202
203     public boolean isalive(int clientno) {
204         if (character[clientno][5] <= 0) {
205             // cleanDied(clientno);
206             return false;
207         }
208         else
209             return true;
210     }
211
212     /** public static void startgame() { //lucianna: ? 這裡有用嗎?
213         //call UDP
214         // System.out.println("position");
215         ul.startUDPBroadCast();
216
217     }*/
218

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219 public void UpdateDirection(int clientno, int Movecode) {
220     int tempx;
221     int tempy;
222     tempx = character[clientno][3];
223     tempy = character[clientno][4];
224     switch (Movecode) {
225         case 0:
226             character[clientno][2] = 0;
227             break;
228         case 1:
229             if (cango(tempx, tempy - 10, 1, clientno)) {
230                 character[clientno][1] = 1;
231                 character[clientno][2] = 1;
232
233                 // drawmap(tempx,tempy,clientno,1);
234                 // System.out.println("UpdateDir " + character[clientno][3] + " " +
235                 // character[clientno][4]);
236             }
237             break;
238         case 3:
239             if (cango(tempx, tempy + 10, 3, clientno)) {
240                 // drawmap(tempx,tempy,clientno,2);
241                 character[clientno][1] = 3;
242                 character[clientno][2] = 1;
243
244                 // System.out.println(character[clientno][3] + " " +
245                 // character[clientno][4]);
246             }
247             break;
248         case 4:
249             if (cango(tempx - 10, tempy, 4, clientno)) {
250                 // drawmap(tempx,tempy,clientno,3);
251                 character[clientno][1] = 4;
252                 character[clientno][2] = 1;
253                 //System.out.println("west dir " +character[clientno][2]);
254                 // System.out.println(character[clientno][3] + " " +
255                 // character[clientno][4]);
256             }
257             break;
258         case 2:
259             if (cango(tempx + 10, tempy, 2, clientno)) {
260                 // drawmap(tempx,tempy,clientno,4);
261                 character[clientno][1] = 2;
262                 character[clientno][2] = 1;
263                 // System.out.println(character[clientno][3] + " " +
264                 // character[clientno][4]);
265             }
266             break;
267     }
268 }

```

//判斷是否下一個想要走的位置可以走, 可以的話回傳true, 否則回傳false

```

272 boolean cango(int tempx, int tempy, int move, int no) {
273     int x1, x2, x3, x4, xx;
274     int y1, y2, y3, y4, yy;
275     int mapx, mapy;
276     x1 = tempx - 30 +1;    //先取出各角落的pixel值
277     x2 = tempx - 10 +1;    //用此角落的值, 去做碰撞的判斷
278     x3 = tempx + 10 -1;    // 加或減一, 是以防碰到邊界, 避免map判斷出錯
279     x4 = tempx + 30 -1;
280     y1 = tempy - 30 +1;
281     y2 = tempy - 10 +1;
282     y3 = tempy + 10 -1;
283     y4 = tempy + 30 -1;
284     if (x1 < 0 || x4 > 600 || y1 < 0 || y4 > 600) { //邊界判定
285         // System.out.println("because of bound");
286         return false;
287     }
288     //map
289     switch (move) {        //根據要移動的方向, 去看map是否有障礙
290         case 1:
291             mapy = y1 / 20;    //map的障礙物, 用array儲存, 所以用/20來取map位置

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```

292         if (map[mapy][x1 / 20] == -1 || map[mapy][x2 / 20] == -1 ||
293             map[mapy][x3 / 20] == -1 || map[mapy][x4 / 20] == -1) {
294             // System.out.println("because of map");
295             return false;
296         }
297         break;
298     case 2: //往2方向的碰撞判定
299         mapx = x4 / 20;
300         if (map[y1/20][mapx] == -1 || map[y2 / 20][mapx] == -1 ||
301             map[y3 / 20][mapx] == -1 || map[y4 / 20][mapx] == -1) {
302             // System.out.println("because of map");
303             return false;
304         }
305         break;
306     case 3: //往3方向的碰撞判定
307         mapy = y4 / 20;
308         if (map[mapy][x1 / 20] == -1 || map[mapy][x2 / 20] == -1 ||
309             map[mapy][x3 / 20] == -1 || map[mapy][x4 / 20] == -1) {
310             // System.out.println("because of map");
311             return false;
312         }
313         break;
314     case 4: //往4方向的碰撞判定
315         mapx = x1 / 20;
316         if (map[y1 / 20][mapx] == -1 || map[y2 / 20][mapx] == -1 ||
317             map[y3 / 20][mapx] == -1 || map[y4 / 20][mapx] == -1) {
318             // System.out.println("because of map");
319             return false;
320         }
321         break;
322     }
323     for (int z = 1; z <= 4; z++) { //去看其他腳色是否有碰撞的情況
324         yy = character[z][4];
325         xx = character[z][3];
326         if (z != no) {
327             if ( (xx - tempx < 60 && xx - tempx > -60) &&
328                 (yy - tempy < 60 && yy - tempy > -60)) {
329                 // System.out.println("because of client" + z);
330                 return false;
331             }
332         }
333     }
334     return true;
335 }
336 }
337

```